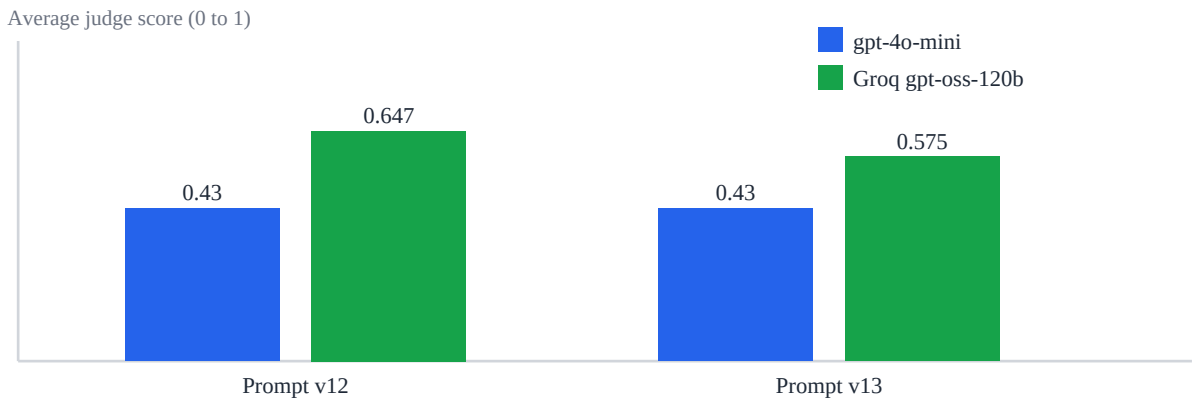
ADK vs Langfuse: evaluation scores

Two ways of testing the same HR agent. ADK checks **how** the agent worked (tools and steps). Langfuse checks **whether the final answer is good** (correct, safe, compliant).

1. Langfuse grid: prompt x model (this session)

Run prefix **claude-test2**. Prompt versions v12 and v13 of *peopleops-root-orchestrator*, two models, 2 golden items (leave balances, apply for earned leave). Judge for every answer: **openai/gpt-4o-mini**. Real calls (mock mode off). Each cell is one run in Langfuse under Datasets > peopleops-golden-benchmark-25 > Runs.

Prompt	gpt-4o-mini (OpenAI)	gpt-oss-120b (Groq)
v12	0.43	**0.647 (best)**
v13	0.43	0.575



How to read it

• The **model** mattered more than the prompt: Groq scored 0.15 to 0.22 higher than gpt-4o-mini on both prompts. Between prompts the gap was small (none on gpt-4o-mini, 0.07 on Groq).
• Only **2 items** were run, so treat this as a smoke test, not a ranking.
• gpt-4o-mini wrote *and* judged its own answers in two cells. Models tend to favour their own output, so Groq's lead is, if anything, understated. A third model as judge would be fairer.

Detail for one cell (v12 + gpt-4o-mini, from an earlier identical run)

Item	Overall	Groundedness	Safety	Helpfulness	Statutory	Latency
Apply 4 days earned leave	0.40	0.0	1.0	0.5	0.0	0.75
Remaining casual + earned leave	0.46	0.0	1.0	0.8	0.0	0.9

Average score 0.43, pass rate 0%, average latency 5.3 s, cost about $0.0009. In that run the agent said it had trouble submitting the leave and used dates starting 23 October instead of 1 December, which points to an agent or tool problem more than a prompt problem. Latency, cost and per-metric detail for the other three cells were not kept.

2. What each side measures

	ADK web (Eval tab)	Langfuse experiments
Question it answers	How did the agent get there? Which tools, which arguments?	Is the answer right, safe and compliant?
What runs	The agent fleet alone, on an isolated test database	The full pipeline: guardrails, PII redaction, memory, agents, output clean-up
Main metrics	Tool-call trajectory (exact match), response match (ROUGE-1 word overlap), semantic match, rubric quality, tool-use quality, hallucinations	Groundedness, helpfulness, statutory compliance, safety / no PII leak, latency, overall
Who scores	Rules for trajectory and ROUGE; LLM judge for the rest	LLM judge for groundedness, helpfulness, statutory; rule + judge for safety; rule for latency
Strength	Catches wrong routing, duplicate or missing tool calls, bad arguments. No judge noise on the exact checks.	Judges meaning. Catches wrong facts and unsafe or non-compliant answers.
Weakness	Exact match punishes harmless rewording. ROUGE rewards word overlap, not correctness.	Cannot see how the answer was made. A wrong tool call can still read fine.
Where to see it	http://localhost:8010/dev-ui, Evals tab	Langfuse > Datasets > Runs

The two ADK eval sets

Set	Cases	Why it exists
peopleops_deterministic	10	The question fixes every tool argument, so an exact tool match is fair. A failure is a real bug. Start here.
peopleops_golden	25	All golden cases. Some need the agent to write free text into a tool argument (a reason, a search phrase, today's date), so rewording can fail the exact check even when behaviour is right.

3. Earlier ADK vs Langfuse comparison (from docs/adk_web_and_langfuse.md)

8 golden cases, model and judge both gpt-4o-mini. ADK ran only trajectory and ROUGE in that run. These figures are taken from the repo doc; they were **not** re-run in this session.

Measure	Result
ADK tool trajectory	0.75 (0.80 on the deterministic cases)
ADK response match (ROUGE-1)	0.18; 12% pass rate at threshold 0.3
Langfuse overall score	0.77; 62% pass rate
Agreement per case	5 Langfuse-only passes, 2 both fail, 1 ADK-only pass
Correlation, Langfuse score vs ROUGE	-0.53 (8 cases, indicative only)

What each side caught that the other missed

• **Only ADK**: *sync_zoho_leave_records* called twice with identical arguments; *apply_leave* given the employee's first name instead of the ID; the balance check skipped before applying for leave. The final answers still read fine.
• **Only Langfuse**: whether a refusal or policy answer is actually correct and safe. ROUGE scored good answers only 0.1 to 0.4, so a low word-match score does not mean a wrong answer.
• **Both fail** is the strongest sign of a real problem.

Not yet done

• The ADK Eval tab has not been run in this session, so there are no fresh ADK scores.
• For both scores on one Langfuse run, call *POST /api/v1/agents/experiments/adk-compare* (HR manager or admin). It runs each case through both and writes the ADK scores onto the Langfuse run.
• ADK and Langfuse numbers use different scales and pipelines, so compare them case by case, not as raw numbers.